

Matt's Timer

Native Android app. Countdown timer + stopwatch, built for back-to-back sets, plus an alarm clock, and it keeps running with the app closed. Installs from dist/MattsTimer.apk - under 700 KB, no dependencies.

It cannot use the internet

There is no INTERNET permission in the manifest, so the phone will not let it reach the network under any circumstances. Nothing loads, nothing syncs, nothing needs signal. The chimes are synthesised on the device at startup rather than shipped as audio files.

The eight permissions it does declare are all local to the phone: VIBRATE, and - so a set keeps running with the app closed - FOREGROUND_SERVICE, FOREGROUND_SERVICE_SPECIAL_USE, POST_NOTIFICATIONS and WAKE_LOCK; and for the alarm clock, accepted on 2026-09-27, USE_EXACT_ALARM (it fires on the minute), RECEIVE_BOOT_COMPLETED (it survives a restart) and USE_FULL_SCREEN_INTENT (it takes the full screen on a locked phone).

The one rule that drives the layout

RESTART puts the clock back to the preset time and runs it immediately. Between back-to-back core sets that is one tap, not two. **RESET**, in the middle, does the same thing but leaves it stopped - it never starts anything.

Presets

35s 45s 1m 5m 10m 15m 20m 30m 60m - nine tiles, three by three.

The app **always launches on 35s**, regardless of what ran last - unless a set is still running, paused or finished, in which case it opens on that set. Presets go dim and inert while the timer is running, so a stray tap mid-set cannot wipe a live timer - pause or let it finish to change duration.

Tap the clock for a one-off time

With the timer stopped - ready, paused or finished - **tap the digits** and two wheels come up, minutes and seconds. Flick them, or tap a number and type it. SET & START applies the value and starts it in one tap, the same path RESTART takes; RESET and RESTART then come back to that value until a preset is tapped. CANCEL or the back button closes it, and the ready line reads TAP TO SET as the reminder. Tapping the clock while it is running, or in the stopwatch, does nothing.

Typing raises a number pad, not the full keyboard, and its check key sets the time and starts it - one key, nothing to reach for afterwards. What you type is read as a number: 5 in the seconds wheel is five seconds, 12 is twelve, and the minutes wheel reads the same way. CANCEL and SET & START stay above the keyboard the whole time it is up: the wheels are what gives way, never the buttons. Closing the picker drops the keyboard with it, and at 0:00 the check key just puts the value in, since there is nothing to start.

The three buttons

Button	Timer	Stopwatch
Left	START / PAUSE / RESUME / GO AGAIN	START / STOP / RESUME
Middle - RESET	Back to the preset time, stopped	Back to zero, clears laps
Right	RESTART - back to the preset time and runs	LAP

Sound

Voice	Character
BELL	C major arpeggio landing on a full triad. Warm, the default
CHIME	Falling G-E-C resolving to an open fifth. Calmer, longer ring
PULSE	Three clipped tones then a high hold. Cuts through a loud gym
MUTE	Silent - vibration still fires

The header button cycles these and always shows the current one; each pick plays a short preview. Changing voice or muting mid-set takes effect on the next cue, with no restart.

At zero

Soft tick at 3, 2, 1 in the selected voice, then the finish chime. The digits go red, the line reads TIME and the phone vibrates. Every tone is a sine with a soft attack and a natural decay, synthesised on the device - no buzzer, no audio files. Cues play on the alarm stream so they carry over gym noise, and vibration toggles separately. Both settings persist.

The screen **flashes for exactly as long as the chime is repeating**. When the repeats end - on their own at two minutes, or because you stopped them - the flash stops with them and the screen sits still on red TIME. **PAUSE on the last instant is a finish**: if the clock has already hit zero by the time the tap lands, the set rings like any other finish rather than freezing at PAUSED 0:00.

The chime repeats, and gets louder

The finish chime does not ring once and give up. It **repeats until you stop it** - RESTART, RESET, a preset, SET & START, the button on the lock screen, tapping either tab, or starting the stopwatch - and each repeat is louder than the last, so a set that ends while you are under a bar does not go unheard. Tapping a tab or starting the stopwatch only silences it: the set stays on TIME and the music stays ducked until RESET or RESTART. Rotating the phone does not silence it.

The ring-out	How it behaves
Volume	30% on the first chime, then 47.5, 65, 82.5, and full from the fifth on
Spacing	the chime's own ring-out plus a one-second gap - every 4 to 5 seconds
It stops itself	nothing new starts more than two minutes past the finish: about 30 chimes on BELL, 25 on CHIME and PULSE

Full means as loud as your alarm volume already is. The app turns its own cue up toward that and no further - it never writes a stream volume, so nothing touches the phone's sliders and nothing else on the phone gets louder. The buzz fires with every chime. On MUTE the buzz carries the sequence alone; with vibration off the chimes carry it alone; with both off it is silent and still ends at two minutes. Changing voice or muting mid-ring takes effect on the next repeat. If the two minutes run out on their own the clock stays on TIME and the music stays ducked - only RESET or RESTART hands that back.

The music ducks under the last three seconds

Three seconds out the app asks the system for transient audio focus, so whatever is playing in the headphones drops under the 3-2-1 and the chime instead of fighting them, and **stays** down at TIME. It comes back up on its own the moment the set is reset or restarted - pausing hands it back too. The app never writes a volume; it only asks for the duck, which is what makes the return gradual and leaves every other app's volume alone. A set the

phone restores inside its last three seconds takes the duck as it comes back.

It keeps running with the app closed

Starting a countdown - or the stopwatch - starts a foreground service, so the set survives leaving the app, the screen going off, and swiping the app out of recents. It sits on the lock screen as an ongoing notification that counts itself down, with the buttons for whatever state it is in: PAUSE and RESTART while running, RESUME and RESTART while paused, RESTART at TIME, STOP for the stopwatch. Tapping the notification opens the app on the live set, and the service stops itself the moment nothing is live.

The notification sits on a **default-importance** channel, so the phone draws it on the lock screen as a card with the countdown and the buttons rather than as a small icon in the top row - which is what the old silent channel got. It stays silent all the same: no sound, no vibration, and it alerts only once, so PAUSE, RESUME and RESTART never pop a banner - every sound comes from the timer itself. On Android 12 and up the card shows immediately instead of ten seconds in. Whether your lock screen draws cards or icons at all is also a phone setting of your own (Samsung: lock-screen notification style), which the app does not touch.

A partial wake lock is held while the countdown runs, and then for the whole ring-out at TIME, so every repeat of the chime and the buzz lands with the screen off; it is released the moment the repeats end. Android 13 and up asks once, on the first start, whether the app may post notifications - the timer runs either way. Every change is written to the phone, so a killed process picks the set back up; a reboot clears it and the app opens fresh on 35s.

Ten minutes and a killed set is dropped. If the process comes back to a set whose finish is more than ten minutes in the past, it is not picked up at all - no red TIME, no notification, no service, no hold on the music. The app is simply fresh on 35s. A set that is still counting comes back however long the process was dead, and one that ended ten minutes ago or less still comes back on a silent TIME. A paused set and the stopwatch are not touched by the rule.

Stopwatch

Second tab in the header. START / STOP, LAP while running, RESET when stopped. Laps list split and cumulative time, newest first. Hundredths resolution. The screen stays awake while anything is counting - and while the chime is still repeating - and releases as soon as that stops; rotating the phone rebuilds the layout without disturbing a live set, or an open wheel picker.

Alarm clock

"Okay lets add in an alarm clock feature next" - Matt, 2026-09-27. Built to docs/ALARM_SPEC.md; **none of this has been run on the phone yet [verify, n=0]**.

Third tab, ALARM. The presets, clock and big buttons give way to a status line (NEXT 5:30 AM · IN 7H 12M, or NO ALARM SET) that moves on the minute, the list of alarms - earliest time of day first, each with its time, ONCE / EVERY DAY / WEEKDAYS / WEEKENDS or its days (M W F), and an ON / OFF switch - a SOUND button and ADD ALARM. Tapping a row opens the editor; the switch only switches. Amber lines appear only when something is wrong: notifications off, full screen off (Android 14 and up), or an alarm that could not be set; the first two open the fix when tapped. Up to 20 alarms.

The editor looks like the timer's picker: an hour wheel 1-12 and a minute wheel 00-59, both typeable, AM / PM, and seven day buttons S M T W T F S - none lit rings once. CANCEL, SAVE, and DELETE for an existing alarm. The number pad's check key only puts the number in; it does not save. After SAVE a line reads RINGS IN 7H 12M for six seconds.

DELETE asks once. The first tap turns it red and reads TAP AGAIN TO DELETE; a second tap within 4 seconds deletes. After 4 seconds, any other tap in the editor, back, or turning the phone, it goes back to plain DELETE. No dialog.

When it rings - app closed, screen off, phone locked - it takes the full screen: ALARM, the time, and one wide STOP. STOP is on the notification too. With the app in front it opens the same screen over the app, and again whenever you come back to the app before the ring ends; what was open underneath stays as it was. No snooze. A one-shot alarm switches itself off; a repeating one moves on to its next day before a note plays.

Sound	Character
DAWN	Four notes climbing slowly, G-C-E-G, settling on a C major chord. The default
HARP	A quick pentatonic ripple up and back down, landing on C and G
TIDE	No melody: two soft chords that swell in and ebb away
LILT	Falling three-note figures, one a second, resting on A and E
BELL / CHIME / PULSE	The timer's own finish chimes

It ignores MUTE and has its own sound, one setting for every alarm; each tap of SOUND plays a preview of three seconds or less. The four new sounds were made to wake without annoying: soft struck tones, nothing above 880 Hz, slow attacks, every note faded out, 4.6 to 6.7 seconds a phrase. **It starts quiet and rises:** 8% on the first phrase, climbing in a straight line to full at 90 seconds, where full is your alarm volume as it already is - no volume slider is ever moved. A phrase repeats 1.5 s after the last ends, with the buzz when the timer's VIB is on.

Fifteen minutes untouched and it stops, leaving a MISSED ALARM card with the time. After a restart, an update, or a clock or time-zone change every alarm is re-armed (5:30 AM stays 5:30 AM local), and after a restart it rings even before the phone is unlocked. An alarm whose time passed while the phone was off rings when the phone comes back if that was 15 minutes ago or less; longer ago it leaves a MISSED ALARM card instead.

The MISSED ALARM card and the status-bar alarm icon open the app on the ALARM tab. That does not silence a timer ring-out - only a tab tap does - and it happens once: turning the phone or coming back later leaves the tab where it is. The timer's lock-screen notification still opens on the live set.

Install on the phone

Easiest: plug the phone into the desktop over USB with USB debugging on, and the install runs from here in one command - no files to move by hand.

By hand: copy `dist/MattsTimer.apk` to the phone over USB, tap it in Files, and allow "install unknown apps" for the file manager the first time. The build is signed with the local debug key, which is fine for sideloading to your own phone.

Rebuilding it

Needs JDK 17 and the Android SDK (compileSdk 35). No AndroidX, no Material library, no third-party dependencies - framework views only, which is why the APK is under 700 KB.

```
cd android && ./gradlew assembleRelease
```

Bumping `versionCode` in `android/app/build.gradle.kts` is only needed if an install ever refuses to replace the previous one.